```
import pandas as pd
import numpy as np
import seaborn as sns
from sklearn.cluster import KMeans
import matplotlib.pyplot as mtp
```

```
data = pd.read_csv('/content/sample_data/Mall_Customers.csv')
data
```

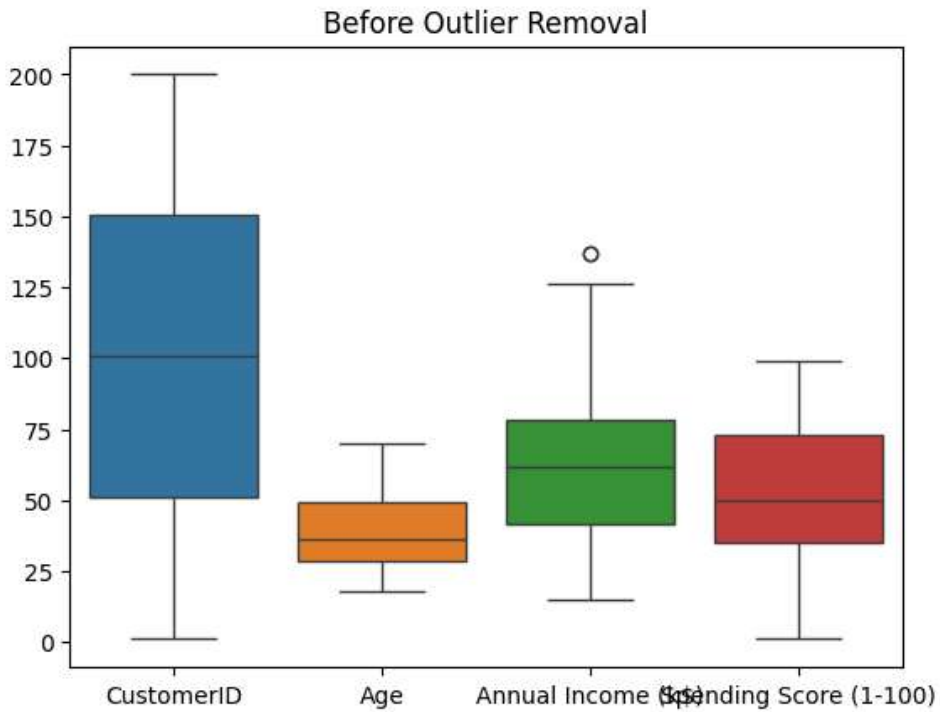
	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
...	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

200 rows × 5 columns

Next steps: New interactive sheet

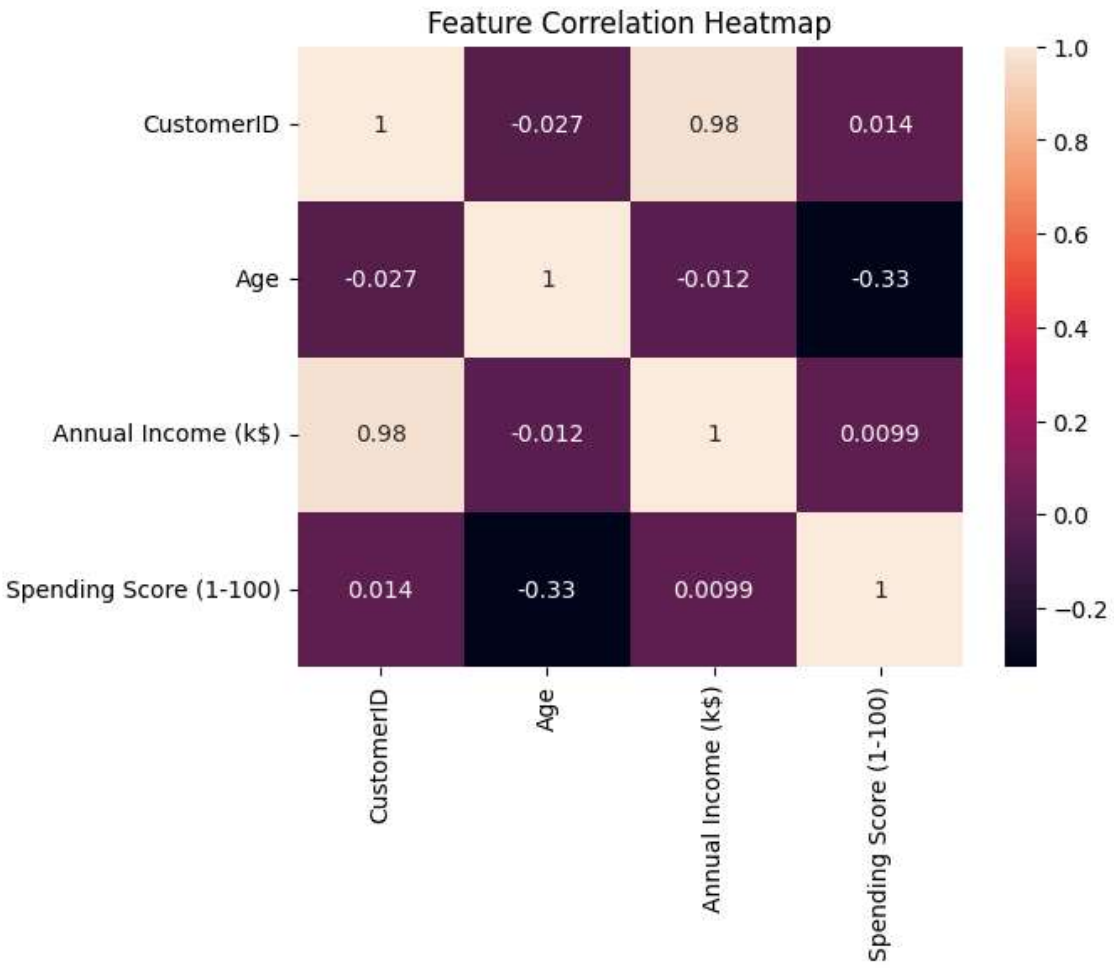
```
numeric_df = data.select_dtypes(include=['number'])


sns.boxplot(data=numeric_df)
plt.title("Before Outlier Removal")
plt.show()
```

```
Q1 = numeric_df.quantile(0.25)
Q3 = numeric_df.quantile(0.75)
IQR = Q3 - Q1

clean_df = numeric_df[~((numeric_df < (Q1 - 1.5 * IQR)) |
                        (numeric_df > (Q3 + 1.5 * IQR))).any(axis=1)]

print("\nAfter removing outliers:")
print(clean_df.shape)
```

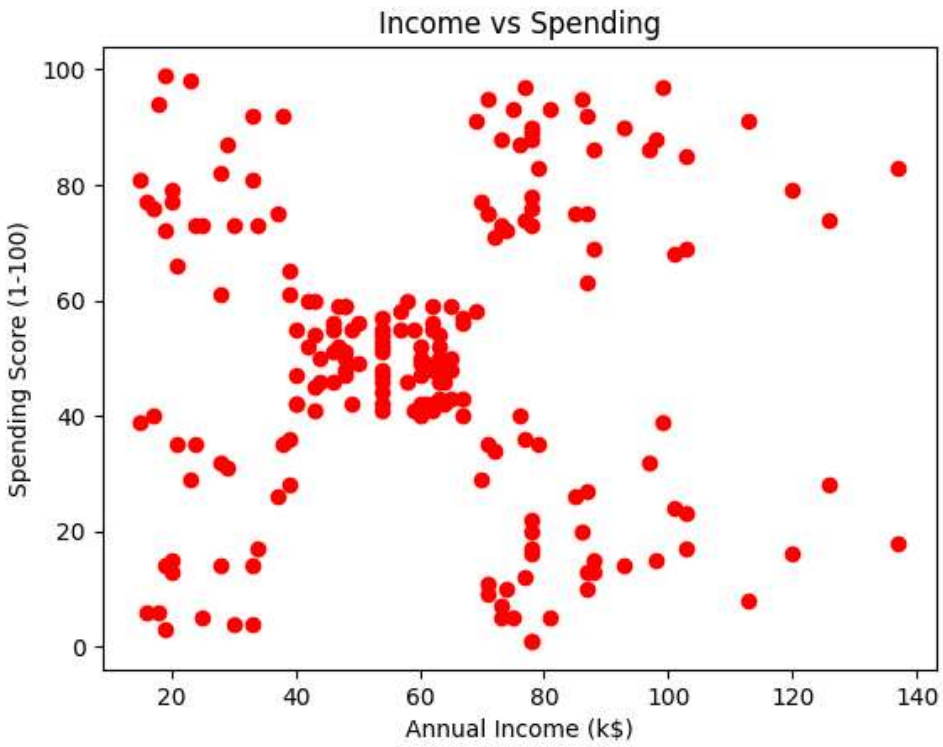



```
After removing outliers:
(198, 4)
```

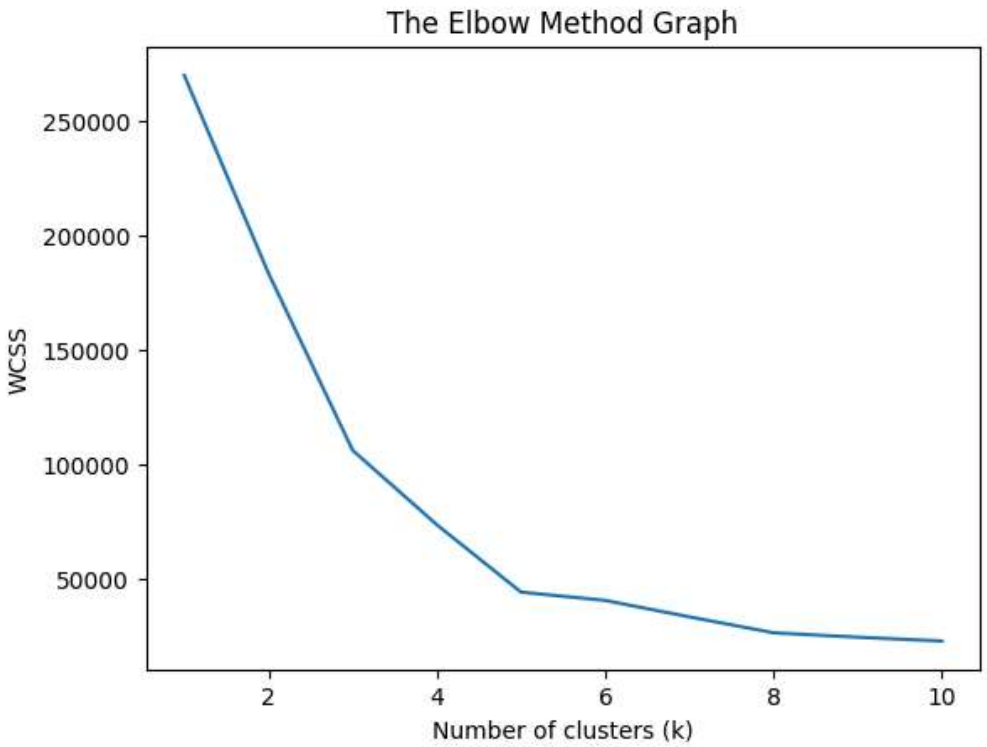
```
corr = data.corr(numeric_only=True)

sns.heatmap(corr, annot=True)
plt.title("Feature Correlation Heatmap")
plt.show()
```



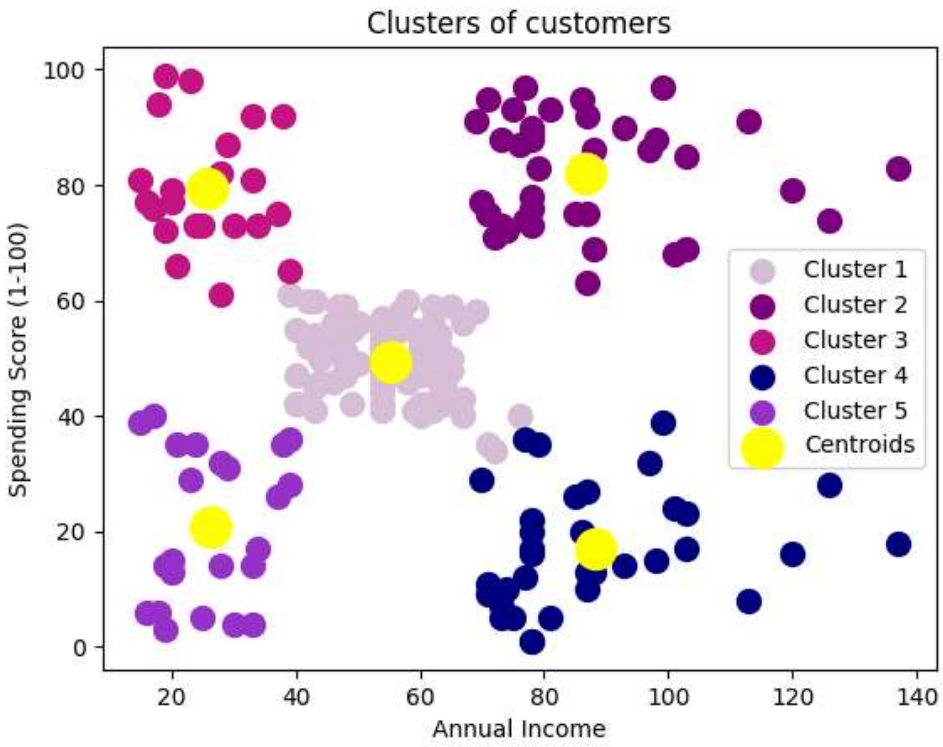

```
plt.scatter(data['Annual Income (k$)'], data['Spending Score (1-100)'], c='red')
plt.title('Income vs Spending')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.show()
```



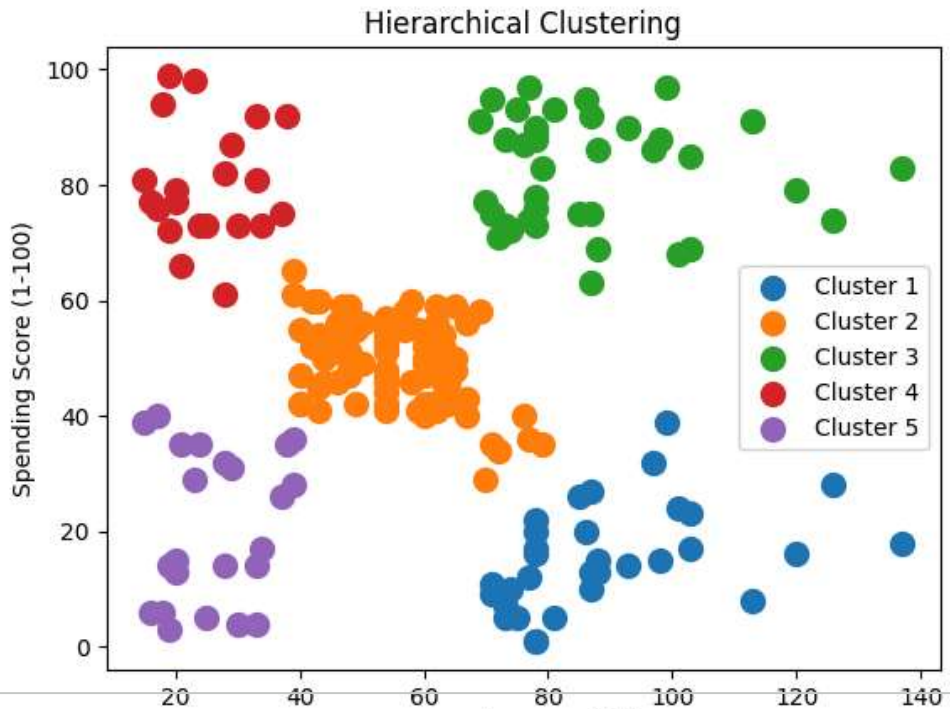

```
x = data.iloc[:, [3,4]].values

wcss_list = []

for i in range(1,11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
    kmeans.fit(x)
    wcss_list.append(kmeans.inertia_)

# plot AFTER loop finishes
mtp.plot(range(1,11), wcss_list)
mtp.title('The Elbow Method Graph')
mtp.xlabel('Number of clusters (k)')
mtp.ylabel('WCSS')
mtp.show()
```




```
kmeans = KMeans(n_clusters = 5, init = 'k-means++', random_state = 42)
y_kmeans = kmeans.fit_predict(x)

plt.scatter(x[y_kmeans == 0, 0], x[y_kmeans == 0, 1], s = 100, c = 'thistle', label = 'Cluster 1')
plt.scatter(x[y_kmeans == 1, 0], x[y_kmeans == 1, 1], s = 100, c = 'purple', label = 'Cluster 2')
plt.scatter(x[y_kmeans == 2, 0], x[y_kmeans == 2, 1], s = 100, c = 'mediumvioletred', label = 'Cluster 3')
plt.scatter(x[y_kmeans == 3, 0], x[y_kmeans == 3, 1], s = 100, c = 'navy', label = 'Cluster 4')
plt.scatter(x[y_kmeans == 4, 0], x[y_kmeans == 4, 1], s = 100, c = 'darkorchid', label = 'Cluster 5')
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s = 300, c = 'yellow', label = 'Centroids')
plt.title('Clusters of customers')
plt.xlabel('Annual Income')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```




```
from sklearn.cluster import AgglomerativeClustering
hc = AgglomerativeClustering(n_clusters=5)
y_hc = hc.fit_predict(x)


plt.scatter(x[y_hc == 0, 0], x[y_hc == 0, 1], s=100, label='Cluster 1')
plt.scatter(x[y_hc == 1, 0], x[y_hc == 1, 1], s=100, label='Cluster 2')
plt.scatter(x[y_hc == 2, 0], x[y_hc == 2, 1], s=100, label='Cluster 3')
plt.scatter(x[y_hc == 3, 0], x[y_hc == 3, 1], s=100, label='Cluster 4')
plt.scatter(x[y_hc == 4, 0], x[y_hc == 4, 1], s=100, label='Cluster 5')

plt.title('Hierarchical Clustering')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```




```
from sklearn.metrics import silhouette_score
score_kmeans = silhouette_score(x, y_kmeans)
score_hc = silhouette_score(x, y_hc)

print("\nSilhouette Score (K-Means):", score_kmeans)
print("Silhouette Score (Hierarchical):", score_hc)
```

```
Silhouette Score (K-Means): 0.553931997444648
Silhouette Score (Hierarchical): 0.5529945955148897
```